

KAIST

2026 Special Topics in Physics Quantum Electronic Devices I:

Interferometers and Fractional Particles

Homework 4

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(1a)

From the unitary condition for the scattering matrix, we can find the following relations:

$$|t|^2 + |r|^2 = 1, \quad (1)$$

$$t^* r + r^* t = 0. \quad (2)$$

Since the state of bosonic particles must be symmetrized, The incoming two-particle state can be written as

$$|\text{incoming}\rangle = \frac{1}{\sqrt{2}} (|\psi_h\rangle |\psi_v\rangle + |\psi_v\rangle |\psi_h\rangle). \quad (3)$$

After the beam splitter, the state becomes

$$|\text{outgoing}\rangle = S \otimes S |\text{incoming}\rangle \quad (4)$$

$$= \frac{1}{\sqrt{2}} (S |\psi_v\rangle S |\psi_h\rangle + S |\psi_h\rangle S |\psi_v\rangle) \quad (5)$$

$$= \frac{1}{\sqrt{2}} (t |\psi_v\rangle + r |\psi_h\rangle) (r |\psi_v\rangle + t |\psi_h\rangle) + \frac{1}{\sqrt{2}} (r |\psi_v\rangle + t |\psi_h\rangle) (t |\psi_v\rangle + r |\psi_h\rangle) \quad (6)$$

$$= \frac{t^2 + r^2}{\sqrt{2}} (|\psi_v\rangle |\psi_h\rangle + |\psi_h\rangle |\psi_v\rangle) + \frac{2tr}{\sqrt{2}} (|\psi_v\rangle |\psi_v\rangle + |\psi_h\rangle |\psi_h\rangle) \quad (7)$$

Hence, the probabilities are:

$$P(2, 0) = 2RT, \quad (8)$$

$$P(0, 2) = 2RT, \quad (9)$$

$$P(1, 1) = (t^2 + r^2)(t^{*2} + r^{*2}) \quad (10)$$

$$= R^2 + T^2 + r^2 t^{*2} + t^2 r^{*2} \quad (11)$$

$$= R^2 + T^2 - 2RT + (r^* t + t^* r)^2 \quad (12)$$

$$= (R - T)^2. \quad (13)$$

(1b)

For fermionic particles, antisymmetrizing the state,

$$|\text{outgoing}\rangle = S \otimes S |\text{incoming}\rangle \quad (14)$$

$$= \frac{1}{\sqrt{2}} (S |\psi_v\rangle S |\psi_h\rangle - S |\psi_h\rangle S |\psi_v\rangle) \quad (15)$$

$$= \frac{1}{\sqrt{2}} (t |\psi_v\rangle + r |\psi_h\rangle) (r |\psi_v\rangle + t |\psi_h\rangle) - \frac{1}{\sqrt{2}} (r |\psi_v\rangle + t |\psi_h\rangle) (t |\psi_v\rangle + r |\psi_h\rangle) \quad (16)$$

$$= \frac{t^2 - r^2}{\sqrt{2}} (|\psi_v\rangle |\psi_h\rangle - |\psi_h\rangle |\psi_v\rangle) \quad (17)$$

Hence, the probabilities are:

$$P(2, 0) = 0, \quad (18)$$

$$P(0, 2) = 0, \quad (19)$$

$$P(1, 1) = (t^2 - r^2)(t^{*2} - r^{*2}) \quad (20)$$

$$= R^2 + T^2 - r^2 t^{*2} - t^2 r^{*2} \quad (21)$$

$$= R^2 + T^2 + 2RT - (r^* t + t^* r)^2 \quad (22)$$

$$= (R + T)^2. \quad (23)$$

(1c)

(1c)

Using second quantization, the problem gives

$$\begin{pmatrix} d_1^\dagger \\ d_2^\dagger \end{pmatrix} = \begin{pmatrix} r & t \\ t & r \end{pmatrix} \begin{pmatrix} d_3^\dagger \\ d_4^\dagger \end{pmatrix} \quad (24)$$

i.e., $d_1^\dagger = r d_3^\dagger + t d_4^\dagger$ and $d_2^\dagger = t d_3^\dagger + r d_4^\dagger$. The incoming state transforms as

$$d_1^\dagger d_2^\dagger |0\rangle = (r d_3^\dagger + t d_4^\dagger)(t d_3^\dagger + r d_4^\dagger) |0\rangle \quad (25)$$

$$= r t (d_3^\dagger)^2 + r^2 d_3^\dagger d_4^\dagger + t^2 d_4^\dagger d_3^\dagger + t r (d_4^\dagger)^2 \quad (26)$$

For the bosonic commutation relation, $[d_i^\dagger, d_j^\dagger] = 0$,

$$d_1^\dagger d_2^\dagger |0\rangle = (t^2 + r^2) d_3^\dagger d_4^\dagger |0\rangle + \sqrt{2} t r \frac{(d_3^\dagger)^2}{\sqrt{2}} |0\rangle + \sqrt{2} t r \frac{(d_4^\dagger)^2}{\sqrt{2}} |0\rangle \quad (27)$$

gives probabilities of $P(2, 0) = P(0, 2) = 2RT$ and $P(1, 1) = (R - T)^2$.

For the fermionic anticommutation relation, $\{d_i^\dagger, d_j^\dagger\} = 0$,

$$d_1^\dagger d_2^\dagger |0\rangle = (t^2 - r^2) d_3^\dagger d_4^\dagger |0\rangle \quad (28)$$

gives probabilities of $P(2, 0) = P(0, 2) = 0$ and $P(1, 1) = 1$.

The results are consistent with the first quantization approach.